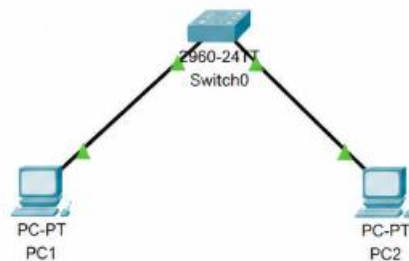


DAY : 6

- ❖ **Switching Basics Theory: Switch operation: MAC address table, frame forwarding. Practical: Observe MAC table population in a simulated network. Task: Explain the process of frame forwarding in switches. Pending**

1. Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

ANS :



```

Switch>enable
Switch#show mac address-table

                        Mac Address Table
                        -----
Vlan    Mac Address      Type    Ports
----    -
Total Mac Addresses for this criterion: 0

Switch#
  
```

Network Created:

- 1 Switch
- PC1
- PC2
- PC1 connected to Switch Port 1
- PC2 connected to Switch Port 2

Initial MAC Address Table

- When the switch is newly connected and no communication has happened yet, the MAC address table is initially empty or contains very few learned entries.
- Command used on the switch : show mac address-table

Observation:

- Initially, the switch does not know which MAC address belongs to which port. It learns MAC addresses only when it receives Ethernet frames from the connected devices.

2. Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table.
Hint: Try pinging in both directions to see how the table updates

ANS :

First, assign IP addresses:

PC1: 192.168.1.1

PC2: 192.168.1.2

Subnet Mask: 255.255.255.0

From PC1, run:

ping 192.168.1.2

After the ping, check the switch:

show mac address-table

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Observation:

The switch now learns the MAC addresses of the PCs.

3. Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

ANS :

Now add PC3 to the switch.

Example:

PC1 → Switch Port 1

PC2 → Switch Port 2

PC3 → Switch Port 3

Assign PC3:

PC3: 192.168.1.3

Subnet Mask: 255.255.255.0

From PC3, ping PC1:

ping 192.168.1.1

Then check the MAC table:

show mac address-table

```
Switch>enable
Switch#show mac address-table

                        Mac Address Table
                        -----
Vlan    Mac Address      Type      Ports
----    -
1       00:11:22:33:44:55  DYNAMIC   Fa0/1
1       00:AA:BB:CC:DD:EE  DYNAMIC   Fa0/2
1       00:CC:CC:CC:CC:CC  DYNAMIC   Fa0/3

Total Mac Addresses for this criterion: 3
```

Observation:

The switch will learn PC3's MAC address on Port 3.

4. Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

ANS :

Given MAC address table:

Port 1 → 00:11:22:33:44:55

Port 2 → 66:77:88:99:AA:BB

A frame arrives at **Port 1** with destination MAC : 66:77:88:99:AA:BB

Step-by-Step Process:

Step 1: The switch receives the frame on **Port 1**.

Step 2: The switch checks the **source MAC address** of the frame and can learn/update which port that source MAC is connected to.

Step 3: The switch looks at the **destination MAC address** : 66:77:88:99:AA:BB

Step 4: The switch searches its MAC address table and finds : 66:77:88:99:AA:BB → Port 2

Step 5: Because the destination is known, the switch forwards the frame **only through Port 2**.

Step 6: The device connected to Port 2 receives the frame and processes it.

5. Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words. Constraint : Do not copy definitions from the text book; use your own understanding and examples.

ANS :

